

AMAL DEV HARIDEVAN

Autonomous Systems Engineer | ROS2, C++, Reinforcement Learning

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SUMMARY

Ph.D. researcher in autonomous systems at York University, specializing in reinforcement learning, high-performance robotic simulation, digital twins, and sim-to-real transfer for UAVs. Built a vectorized Gazebo framework exceeding 100,000 simulation steps per second, deployed flight-control policies with sub-1 ms inference on Jetson hardware, and contributed upstream fixes to PyTorch and Gazebo. Author of four peer-reviewed publications spanning ICRA, IROS, ICUAS, and IEEE Transactions on Instrumentation and Measurement.

EXPERIENCE

SDCNLab – Graduate Autonomous Systems and Robot Learning Researcher – Toronto, Ontario

May 2021 – Present

- Re-engineered Gazebo's IPC-based execution model into a scalable synchronous simulation framework for vectorized drone reinforcement-learning environments, achieving 100,000+ simulation steps/sec on CPUs, 3× the throughput of NVIDIA Isaac Sim on GPU in internal benchmarks.
- Built a hardware-accelerated inference pipeline using ONNX and TensorRT to deploy PPO-based flight-control policies on a QDrone2 Jetson Xavier edge computer, achieving sub-1 ms inference latency for real-time onboard control.
- Designed and implemented an EKF-based full-state estimator fusing IMU, camera-derived ORB visual features, and ranging data for real-time autonomous quadrotor flight.
- Co-developed and deployed Ghost-DeblurGAN, a lightweight GhostNet-based model restoring motion-blurred fiducial markers at 30 Hz on QCar 1 onboard processor, improving marker detection accuracy by 27% for real-time robotic localization.

York University – Teaching Assistant – Toronto, Ontario

May 2023 – Present

- Led laboratory sessions and graded assessments for ENG 4550: Introduction to Control Systems, helping students apply control theory, interpret experimental results, and communicate technical findings.

Quanser – Autonomous Systems Engineering Intern – Markham, Ontario

Aug 2023 – Mar 2025

- Developed and deployed C++ ROS1 and ROS2 interfaces for camera, IMU, motor commands, state estimates, and trajectory planning, enabling researchers to integrate the QDrone2 platform into ROS-based autonomy workflows.
- Architected a Gazebo Harmonic digital twin environment replicating QDrone2 vehicle dynamics, onboard sensors, and controller behaviour, enabling safe prototyping and validation before physical flight testing.
- Implemented a configurable optical-flow tracking pipeline as a C++ OpenCV S-function for MATLAB/Simulink on ARM platforms, overcoming the limited customization of standard MATLAB blocks and enabling more advanced real-time embedded vision applications.

OPEN-SOURCE CONTRIBUTIONS

[\[github.com/pytorch/pytorch/pull/166554\]](https://github.com/pytorch/pytorch/pull/166554) – PyTorch Core Contributor – TorchDynamo

Nov 2025

- Diagnosed and fixed a TorchDynamo graph-caching bug treating 0-D tensor values as compile-time constants; implemented a tensor-aware decomposition and regression tests, restoring correct dynamic behaviour in torch.compile and landing the fix upstream.

[\[github.com/gazebo/gz-sim/pull/2618\]](https://github.com/gazebo/gz-sim/pull/2618) – Gazebo Simulator Contributor – gz-sim

Feb 2025

- Resolved a Python GIL ownership bug in Gazebo's C++ PythonSystemLoader by introducing scoped GIL acquisition around PreUpdate, Update, and PostUpdate callbacks, enabling reliable execution of embedded Python system plugins; added regression tests and merged the fix upstream.

SKILLS

Languages: Python, C++, MATLAB

Framework & Libraries: PyTorch, OpenCV, ROS1, ROS2, Pybind11, ONNX, TensorRT

Simulation & Robotics: Gazebo Harmonic, Isaac Sim, Simulink, Digital Twins

Computer Vision & ML: Reinforcement Learning (PPO), EKF, Motion Planning, Visual Odometry (ORB)

Development Tools: Git, Docker, CMake, GDB, Linux (Ubuntu), Google Cloud.

EDUCATION

York University – Ph.D. in Autonomous Systems and Digital Twins – Toronto, Canada

2025 – 2029 (expected)

- Researching efficient sim-to-real and domain-adaptation methods for autonomous UAV control using reinforcement learning, vision-language models, and high-fidelity digital twins.
- Developing scalable training and evaluation pipelines for learned quadrotor controllers under control delays, wind disturbances, and uncertain vehicle dynamics.
- Demonstrated zero-shot transfer of reinforcement-learning policies from Gazebo simulation to physical quadrotors, achieving a 95% target-tracking success rate without real-world fine-tuning.

York University – MSc in Autonomous Systems and Digital Twins – Toronto, Canada

2023 – 2025

- Award: MITACS Accelerate Grant (\$10,000) for industry-partnered research.
- Developed a digital twin framework for quadrotors to prototype and validate non-linear control, state estimation (EKF), and Deep RL pipelines.

Lassonde School of Engineering, York University – Honours Bachelor of Engineering, Space Engineering – Toronto, Canada

2018 – 2022

- GPA: 3.6/4.00
- Lassonde Undergraduate Student Research Award (\$6,000)

PUBLICATIONS

- Rezaei, S., Kang, J., Haridevan, A., and Shan, J., 2026. SEP-NMPC: Safety Enhanced Passivity-Based Nonlinear Model Predictive Control for a UAV Slung Payload System. In 2026 IEEE International Conference on Robotics and Automation (ICRA).
- Liu, Y., Shan, J., Haridevan, A. and Zhang, S., 2024. L-PR: Exploiting LiDAR Fiducial Marker for Unordered Low Overlap Multiview Point Cloud Registration. IEEE Transactions on Instrumentation and Measurement. (github.com/yorklyb/L-PR)
- Haridevan, A.D. Kang, J., Yuan, M. and Shan, J., 2024, June. ROS2-Gazebo Simulator for Drone Applications. In 2024 International Conference on Unmanned Aircraft Systems (ICUAS) (pp. 1232-1238). IEEE.
- Liu, Y., Haridevan, A., Schofield, H. and Shan, J., 2022, October. Application of Ghost-DeblurGAN to fiducial marker detection. In 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 6827-6832). IEEE. (github.com/York-SDCNLab/Ghost-DeblurGAN)

AWARDS

- Second Prize, American Control Conference Self-Driving Car Competition — Jul 2024